

PYALITSA, Russian Federation

WMO: 223490

Lat: 66.189N

Lon: 39.524E

Elev: 28

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.8	-5.3	-14.3	2.5	-9.3	-9.5	3.3	-4.8	36.0	20.0	32.9	20.9	6.7	330	0.872

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	64.9	59.0	61.5	57.0	58.7	55.3	60.1	63.3	57.7	60.5	55.8	58.1	11.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.3	72.8	62.2	56.2	67.3	59.3	54.4	63.1	57.2	26.5	63.3	24.9	60.6	23.7	58.3	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.4	27.7	24.7	-14.5	70.9	5.8	4.0	-18.7	73.9	-22.1	76.2	-25.4	78.5	-29.6	81.4		
			-15.6	64.1	6.1	3.2	-19.9	66.5	-23.5	68.3	-26.9	70.2	-31.3	72.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	15.5	15.1	18.8	27.5	36.9	45.5	51.9	50.1	45.1	36.1	27.3	22.5	(6)
(7)		DBStd	14.70	9.88	10.92	8.57	6.33	6.30	5.80	5.49	4.29	4.62	5.76	6.97	8.50	(7)
(8)		HDD50	6474	1070	977	967	675	410	161	42	51	157	432	680	852	(8)
(9)		HDD65	11758	1535	1397	1432	1125	873	586	408	461	597	897	1130	1317	(9)
(10)		CDD50	193	0	0	0	0	3	26	100	55	9	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	11.6	13.5	12.5	11.7	10.7	10.9	10.6	9.7	9.2	10.5	12.4	13.6	14.4	(14)
(15)	Precipitation	PrecAvg	17.7	1.0	0.8	0.8	0.7	1.5	1.5	2.0	2.3	2.0	2.3	1.5	1.3	(15)
(16)		PrecMax	25.1	1.7	1.7	1.4	1.3	2.8	3.9	5.0	5.2	4.2	4.8	3.7	2.4	(16)
(17)		PrecMin	12.8	0.3	0.3	0.2	0.2	0.5	0.4	0.4	0.7	0.5	0.8	0.3	0.6	(17)
(18)		PrecStd	2.7	0.4	0.4	0.4	0.4	0.7	0.8	1.1	1.3	1.0	1.1	0.7	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	35.2	36.5	43.3	59.3	65.4	71.4	66.1	58.5	48.6	41.8	36.7	(19)
(20)		MCWB	32.1	33.8	33.4	39.1	50.8	57.6	63.5	60.0	53.5	47.0	40.6	34.8	(20)	
(21)		2%	DB	32.0	32.3	33.3	39.3	52.3	60.8	66.8	61.2	55.2	46.9	39.2	35.3	(21)
(22)		MCWB	31.4	31.5	31.7	36.0	46.7	54.4	60.6	57.5	52.1	45.5	38.1	34.0	(22)	
(23)		5%	DB	30.9	30.7	32.0	37.0	48.9	57.3	63.4	58.6	53.4	45.2	37.4	34.3	(23)
(24)		MCWB	30.3	29.8	31.1	34.3	44.4	52.7	58.6	56.1	51.6	43.8	36.4	33.2	(24)	
(25)		10%	DB	29.0	28.7	30.7	35.1	46.0	54.4	60.5	56.5	52.1	43.8	36.0	33.1	(25)
(26)		MCWB	28.2	27.9	29.8	33.1	42.7	50.6	56.9	54.3	50.4	42.2	34.8	32.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	34.0	33.8	39.3	52.5	58.7	64.8	61.3	55.2	47.8	41.1	35.4	(27)
(28)		MADB	32.8	35.0	35.5	42.6	58.7	62.4	69.9	64.2	57.6	48.2	41.8	35.9	(28)	
(29)		2%	WB	31.5	31.7	32.2	36.3	47.4	55.3	61.6	58.2	53.0	45.8	38.4	34.3	(29)
(30)		MADB	31.8	32.1	33.0	38.7	51.5	59.1	65.5	60.6	54.1	46.6	39.0	35.0	(30)	
(31)		5%	WB	30.3	30.0	31.2	34.6	44.9	53.2	59.3	56.3	51.9	43.9	36.5	33.4	(31)
(32)		MADB	30.8	30.7	31.9	36.3	48.0	57.1	62.6	58.3	53.1	45.0	37.2	34.1	(32)	
(33)		10%	WB	28.3	28.0	29.9	33.4	43.0	51.0	57.1	54.8	50.7	42.4	34.9	32.2	(33)
(34)		MADB	29.0	28.7	30.7	34.7	46.1	54.1	60.0	56.2	51.9	43.7	35.8	32.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.0	9.7	11.2	9.7	9.3	10.3	10.1	9.1	8.9	7.1	6.8	7.7	(35)	
(36)		5% DB	MCDBR	6.9	8.4	8.5	9.9	14.6	15.0	14.5	12.0	10.1	6.8	6.0	5.2	(36)
(37)		MCWBR	7.0	8.4	7.9	7.7	10.0	10.4	9.9	8.6	7.9	6.1	6.1	5.4	(37)	
(38)		5% WB	MCDBR	6.9	8.4	8.0	8.7	13.3	13.6	13.1	10.8	9.4	6.2	6.1	5.2	(38)
(39)		MCWBR	7.1	8.4	7.6	7.2	9.9	10.0	9.8	8.3	7.8	5.9	6.3	5.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.222	0.275	0.306	0.335	0.347	0.342	0.374	0.350	0.328	0.299	0.357	0.366	(40)	
(41)		taud	1.938	2.054	2.109	2.181	2.398	2.499	2.434	2.476	2.507	2.299	2.561	2.565	(41)	
(42)		Ebn at Noon	61	176	234	260	271	276	261	257	236	174	64	15	(42)	
(43)		Edh at Noon	6	18	29	35	31	29	30	26	20	15	6	3	(43)	
(44)	All-Sky Solar Radiation	RadAvg	22	162	626	1192	1518	1715	1614	1099	587	197	38	3	(44)	
(45)		RadStd	3	30	74	91	126	161	170	112	57	34	5	0	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	+1.13	+4.50	+4.27	N/A	N/A	N/A	-284	-347	N/A	N/A	(47)

Nomenclature: See separate page